

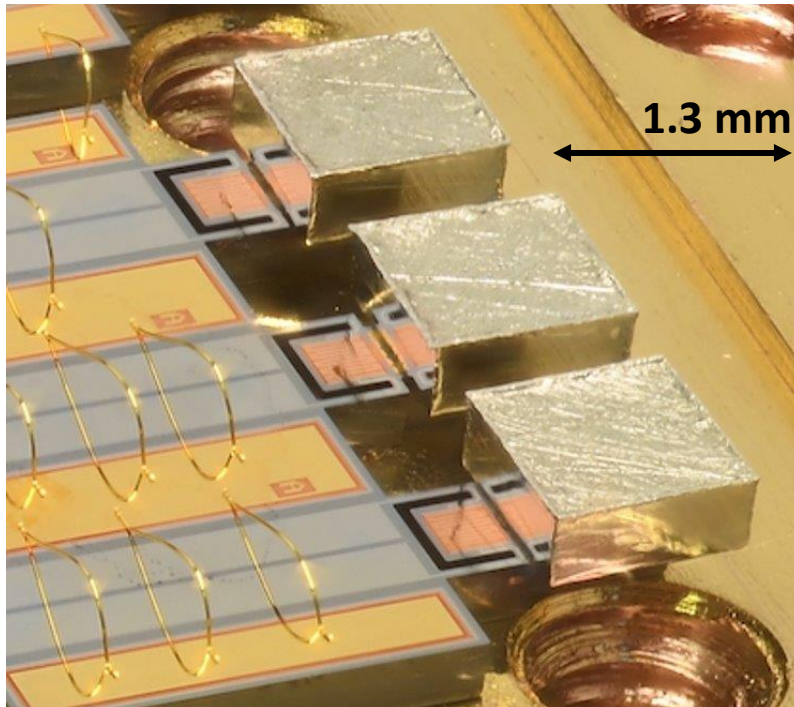
Achieving 1 % Assay of Special Nuclear Materials in 2 Minutes with Microcalorimeter-Array Gamma-Ray Spectroscopy



University of Colorado
Boulder



Dan Becker
University of Colorado



Enabling near-real-time monitoring of
nuclear materials ... via high-resolution
gamma-ray spectroscopy

ARPA-E Fission Annual Meeting / Oct 1 - 2, 2025

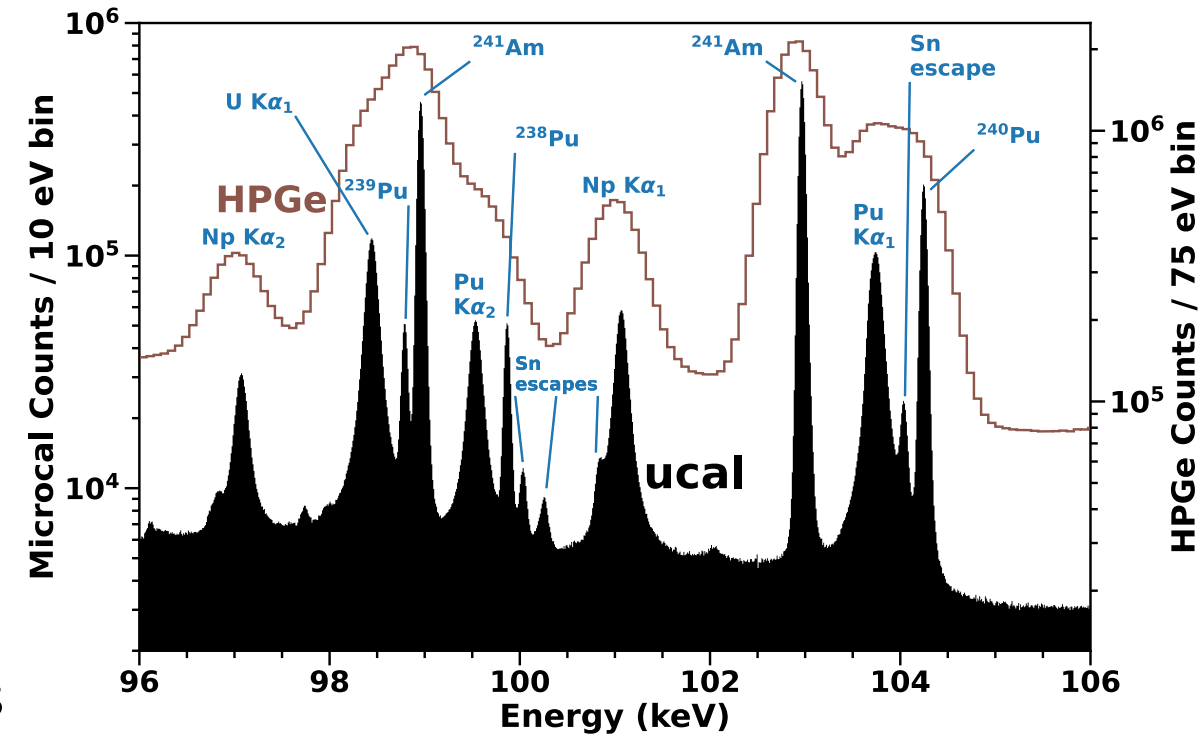
Project Objective and Impact on Nuclear Materials Control and Accounting

Project Objectives

- Enable assay of complex Special Nuclear Materials (SNM) mixtures to within **1 % accuracy** in **2 minutes**
- Enable reductions in CapEx and OpEx costs for nuclear fuel recycling facilities

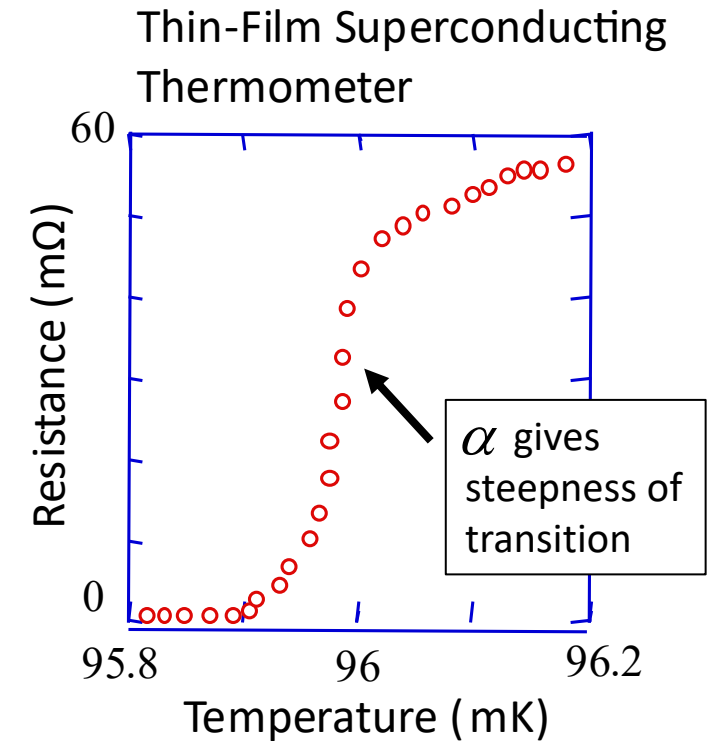
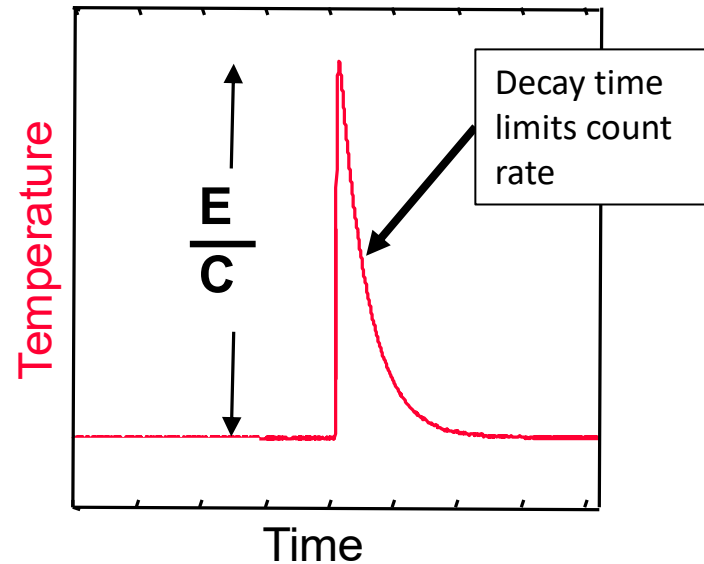
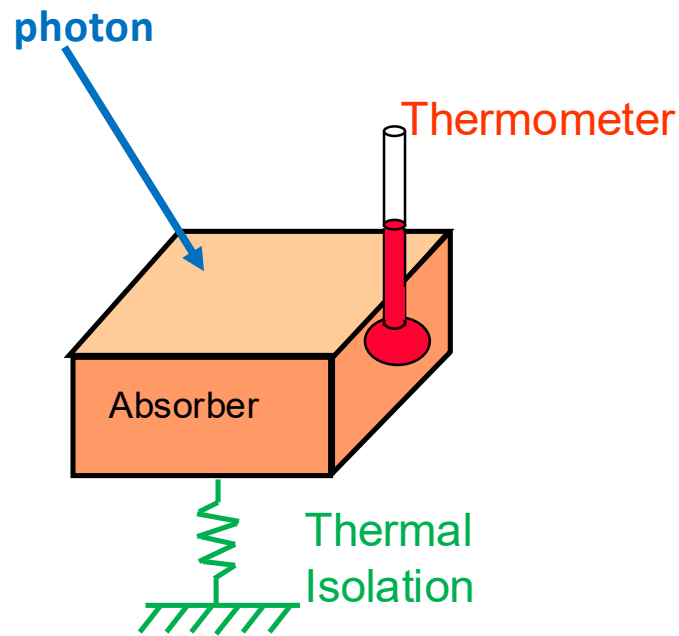
Core Enabling Technology

- Low-temperature microcalorimeters
- Superior resolution of microcalorimeter arrays enables separation of closely spaced gamma peaks, enabling Pu isotopic assay to within 1 %
- Our work will make these detectors fast, enabling high statistics measurements in 2 min



Unique combination of energy
resolution and collection efficiency

Superconducting Microcalorimeters



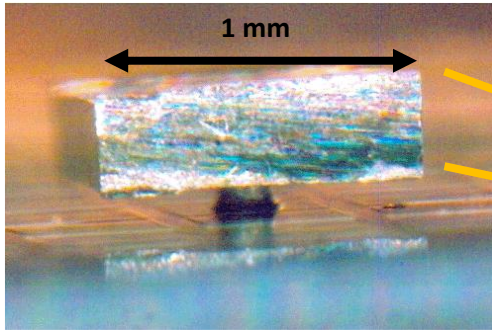
$$\Delta E_{TES} \sim \sqrt{\frac{4k_B T^2 C}{\alpha}} \Rightarrow \begin{matrix} T, C & \text{small} \\ \alpha & \text{large} \end{matrix}$$

Operating at low temperatures gives $E/\Delta E > 1000$
from 1 keV to 5 MeV

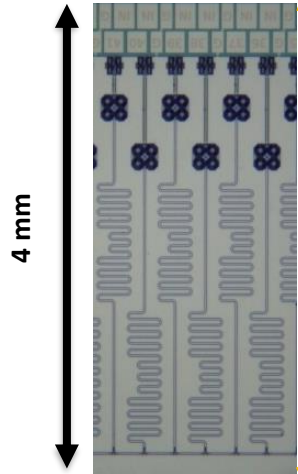
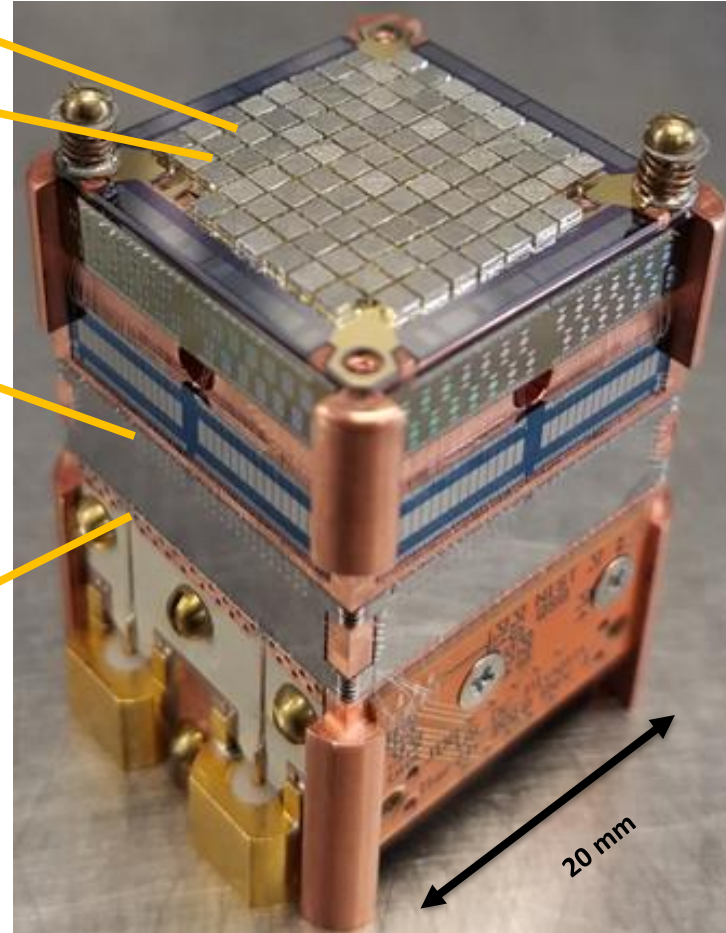
Critical Design Tasks: thermal engineering, choice of absorber material, attachment process, T_c

Physical Realization

Microcalorimeter pixel



Microcalorimeter array subassembly
operates at < 100 mK



Readout: cryogenic
SQUID multiplexor chip

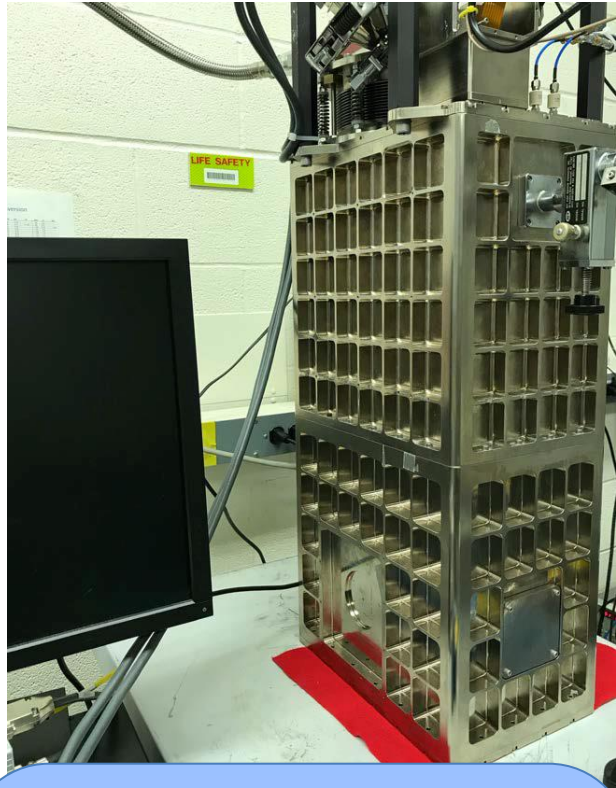
Leverages fundamental quantum
behavior of superconductors for
detection and readout

Commercial Refrigerator:
Cryogen-free, reliable



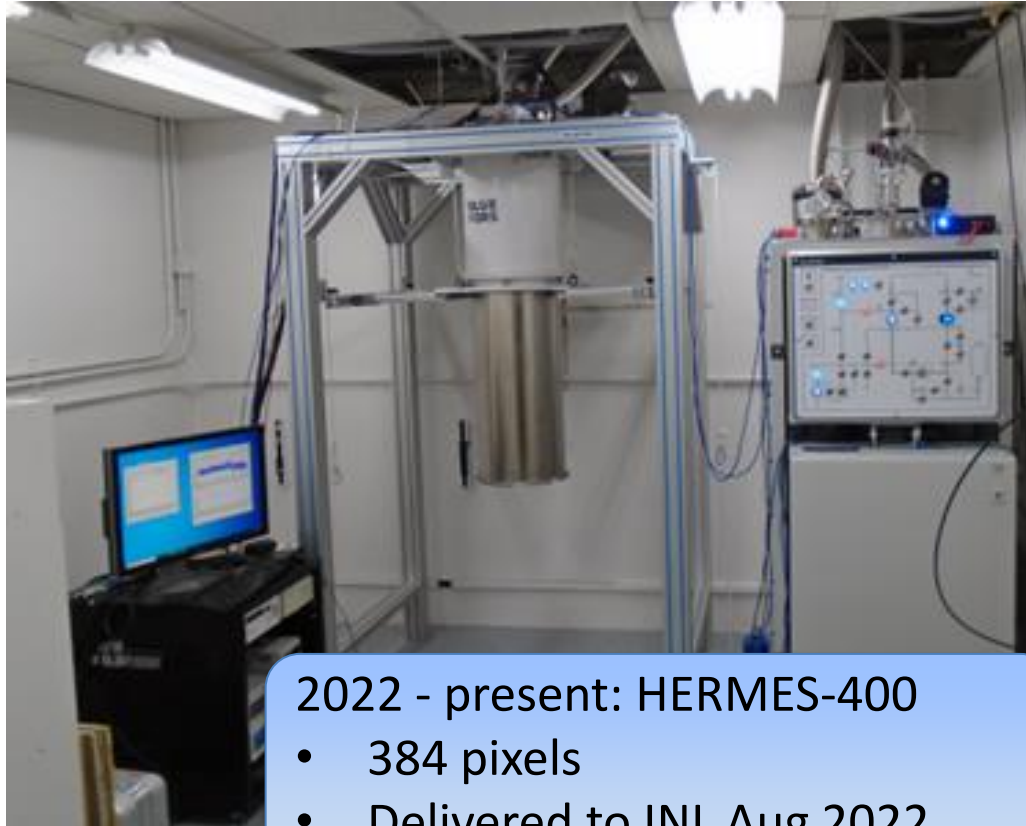
Detectors view samples through a
window in the cryostat. In this case the
window faces down, but sideways-facing
windows are also used

Deployed Microcalorimeter Spectrometers



2019 - present: SOFIA spectrometer

- 256 pixels, portable
- Commissioned at LANL in 2019
- Measurements at PF-4
- Currently at ORNL

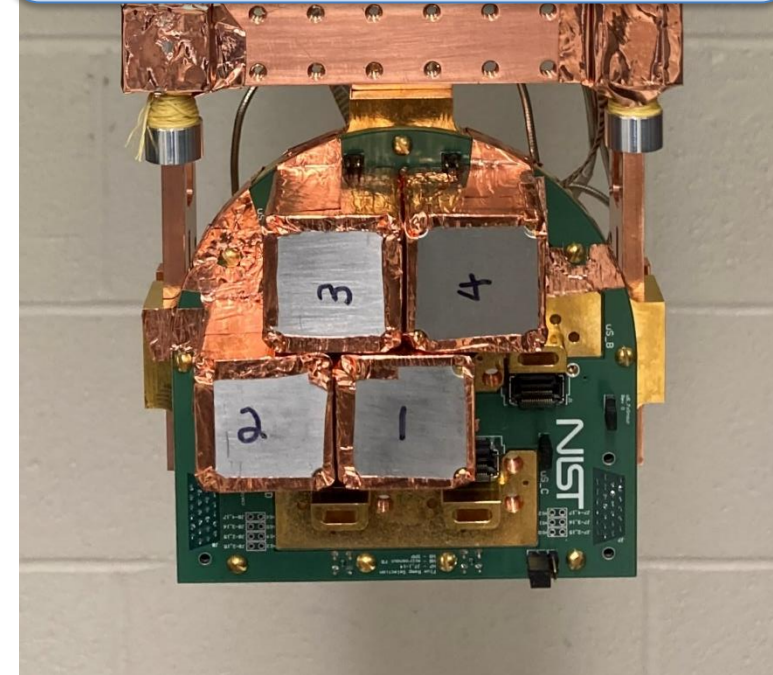


2022 - present: HERMES-400

- 384 pixels
- Delivered to INL Aug 2022

Sep 2024: HERMES-700

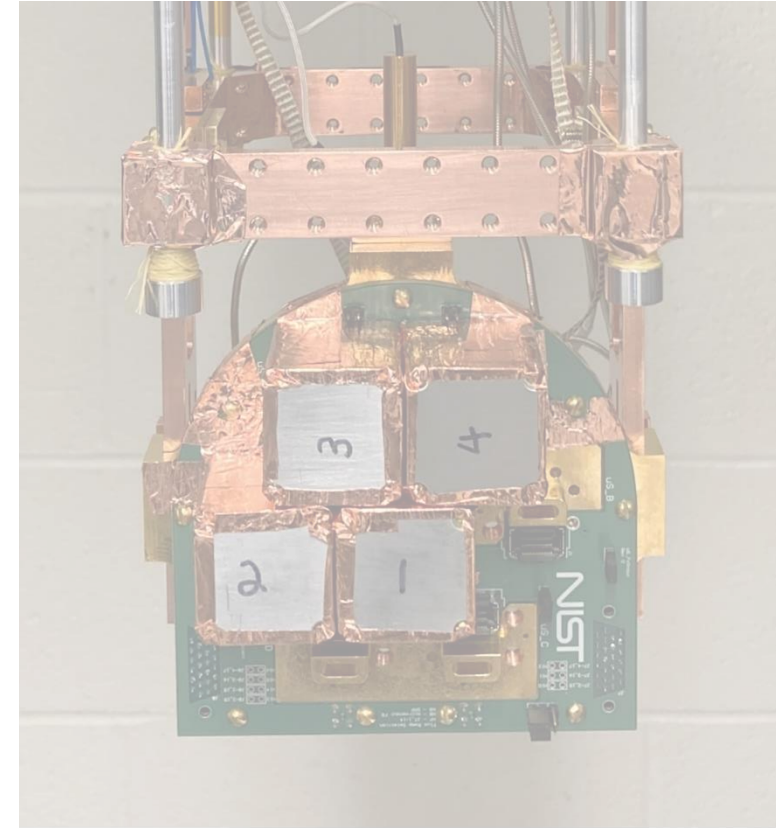
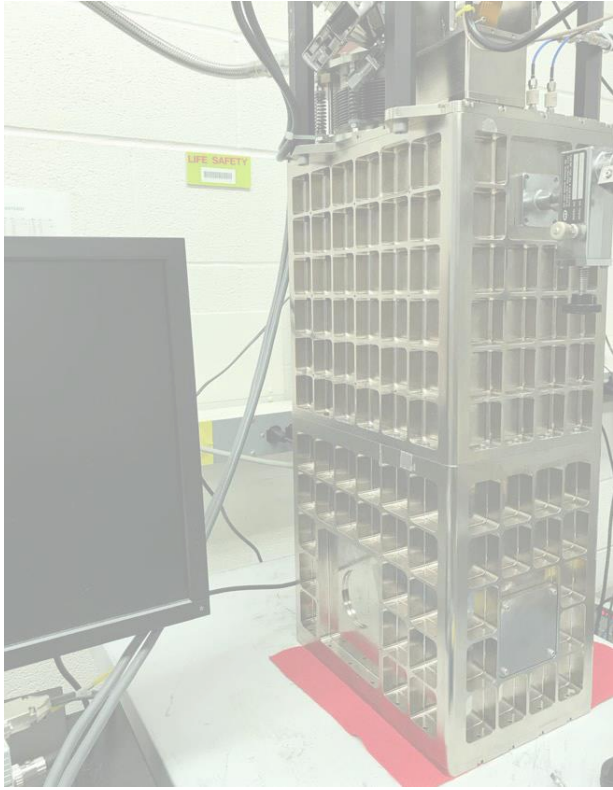
- Up to 672 pixels
- PNNL



Planned instruments in FY26+

- 3 SOFIA-style instruments (IAEA, PF-4, ...)
- Exotic Atom Spectroscopy
- Gamma-ray, hard x-ray astronomy (proposed)

Deployed Microcalorimeter Spectrometers



✓ Existing microcalorimeter spectrometers work well!

☹ But acquisition times are on the scale of ~10 hours

Project Goals



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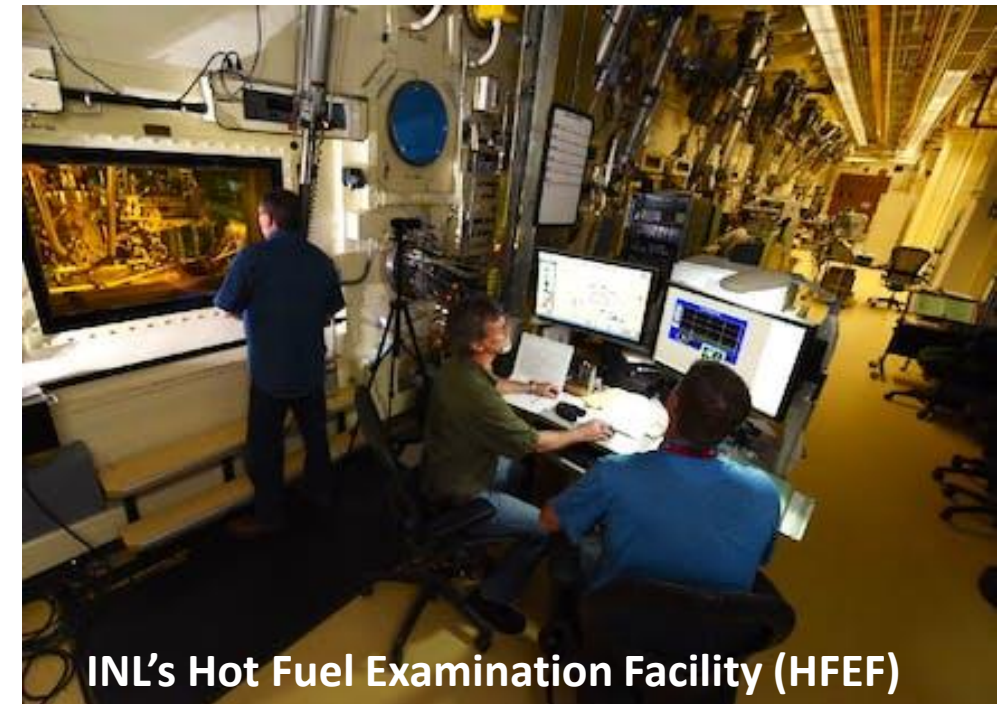
ARPA-E CURIE Program Goals:

- Make 10x reduction in cost of nuclear fuel recycling
- Includes goal of measurements in 2 minutes

Key Technical Project Goals

- 20x increase in TES microcal speed to enable Near Real Time Analysis of nuclear materials
- Demonstrate performance on relevant samples from INL's HFEF

Spectrometer Metric	State of the Art	Proposed
Energy Resolution	< 100 eV (at 100 keV)	unchanged
Per-Detector Count Rates	10 – 20 cps	200 cps
τ_1 % (time to fall to 1 % of peak)	16 ms	750 us
Athermal Tails	Significant	Negligible



INL's Hot Fuel Examination Facility (HFEF)

Project Team



Project PI:
Dan Becker



**Quantum Sensors Group
at CU and NIST**

CU and NIST work together
on core technology
development and T2M tasks



NIST Co-PI:
Dan Schmidt



LANL provides gamma-
spectroscopy expertise and testing
resources (both lab and personnel)



LANL Co-PI:
Rico Schoenemann

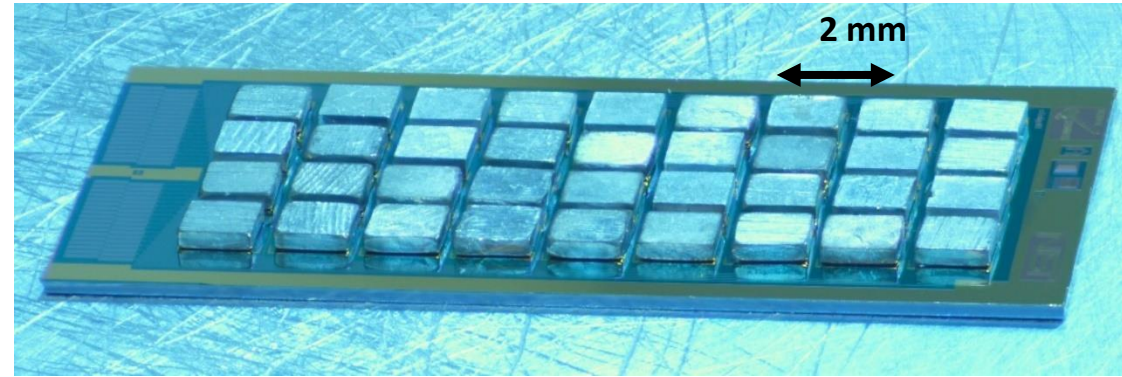
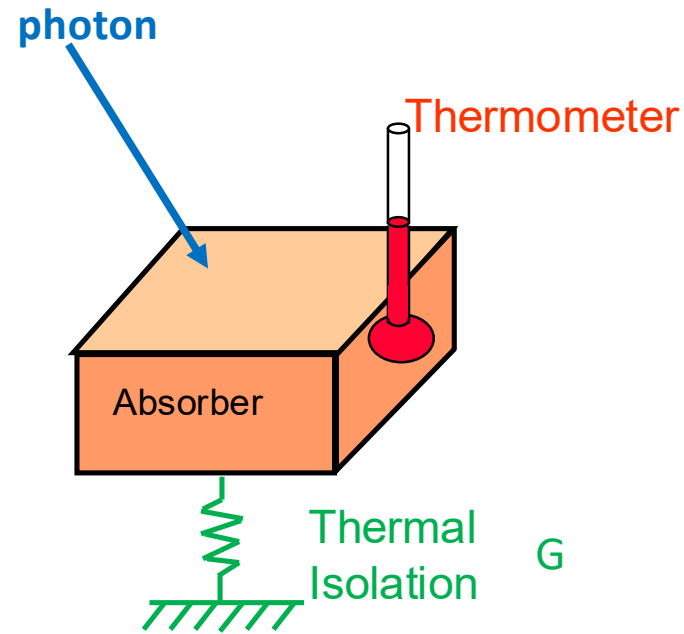


INL provides pyroprocessing
samples and hosts the INL
spectrometers which will be used
for validation of our designs

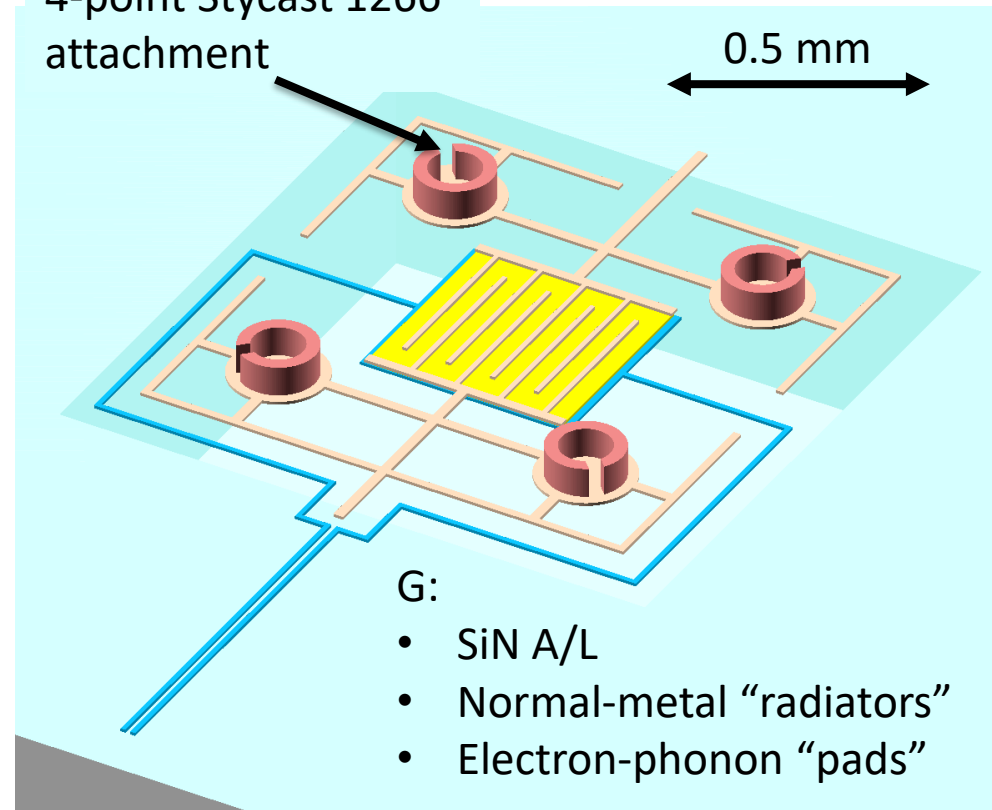
INL Co-PI:
Ammon Williams



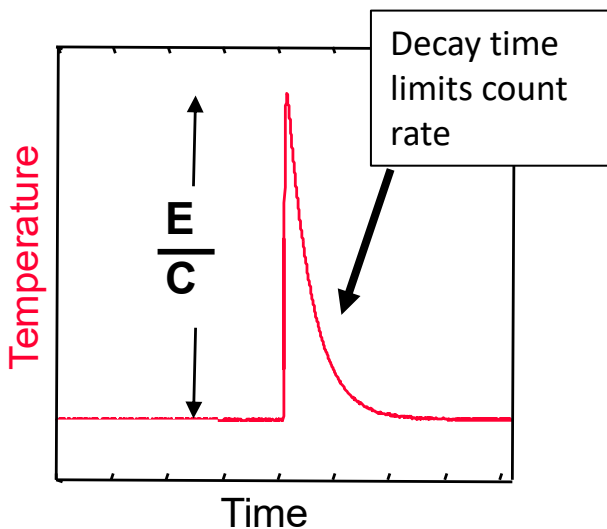
Pre-CURIE Detector Architecture: Membranes



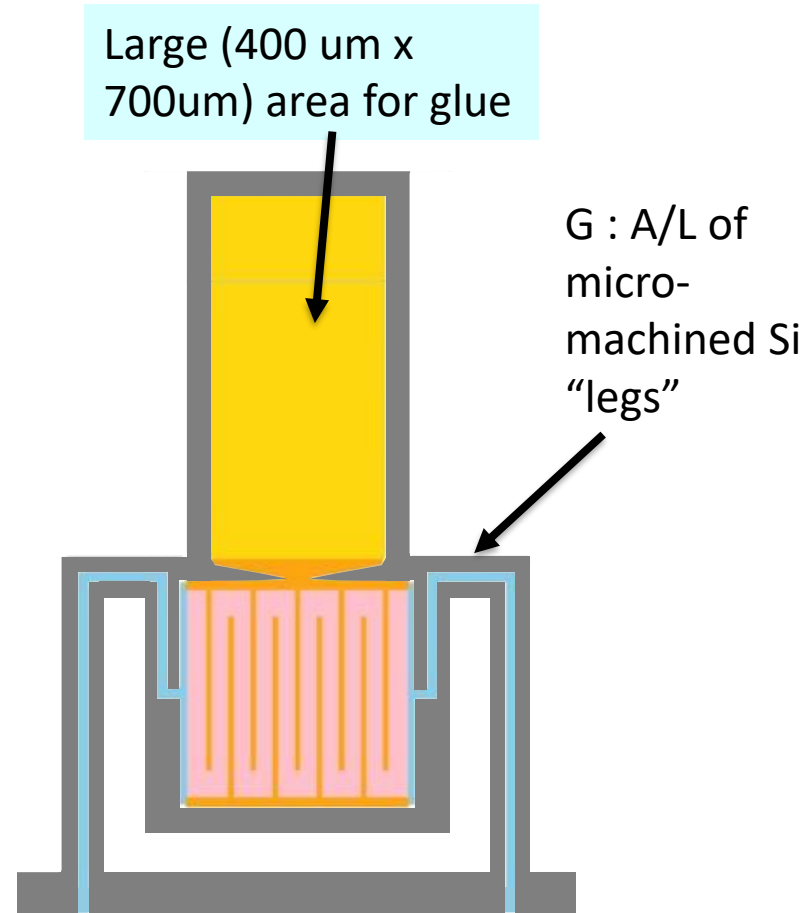
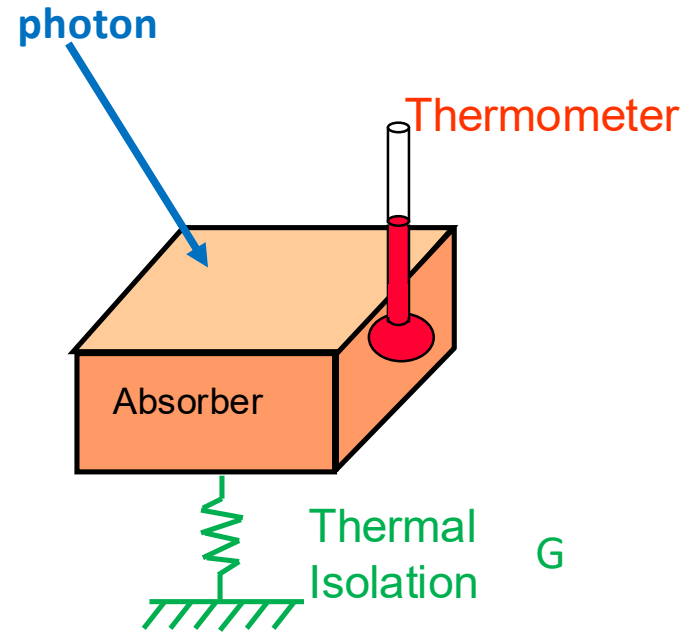
4-point Stycast 1266
attachment



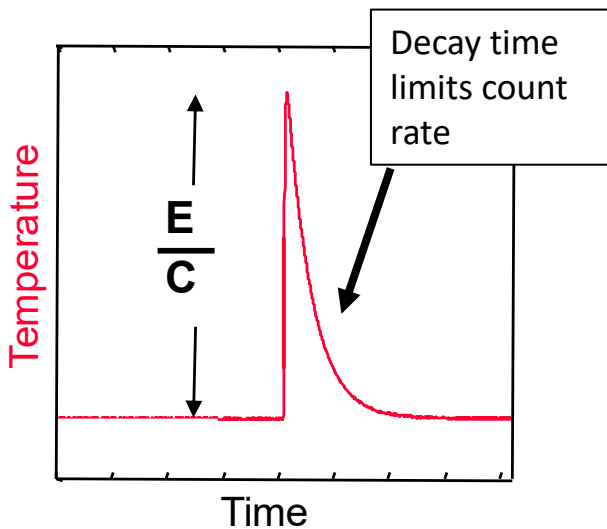
☹️ Finicky Assembly
☹️ Fragile
☹️ Limited ability to
increase G



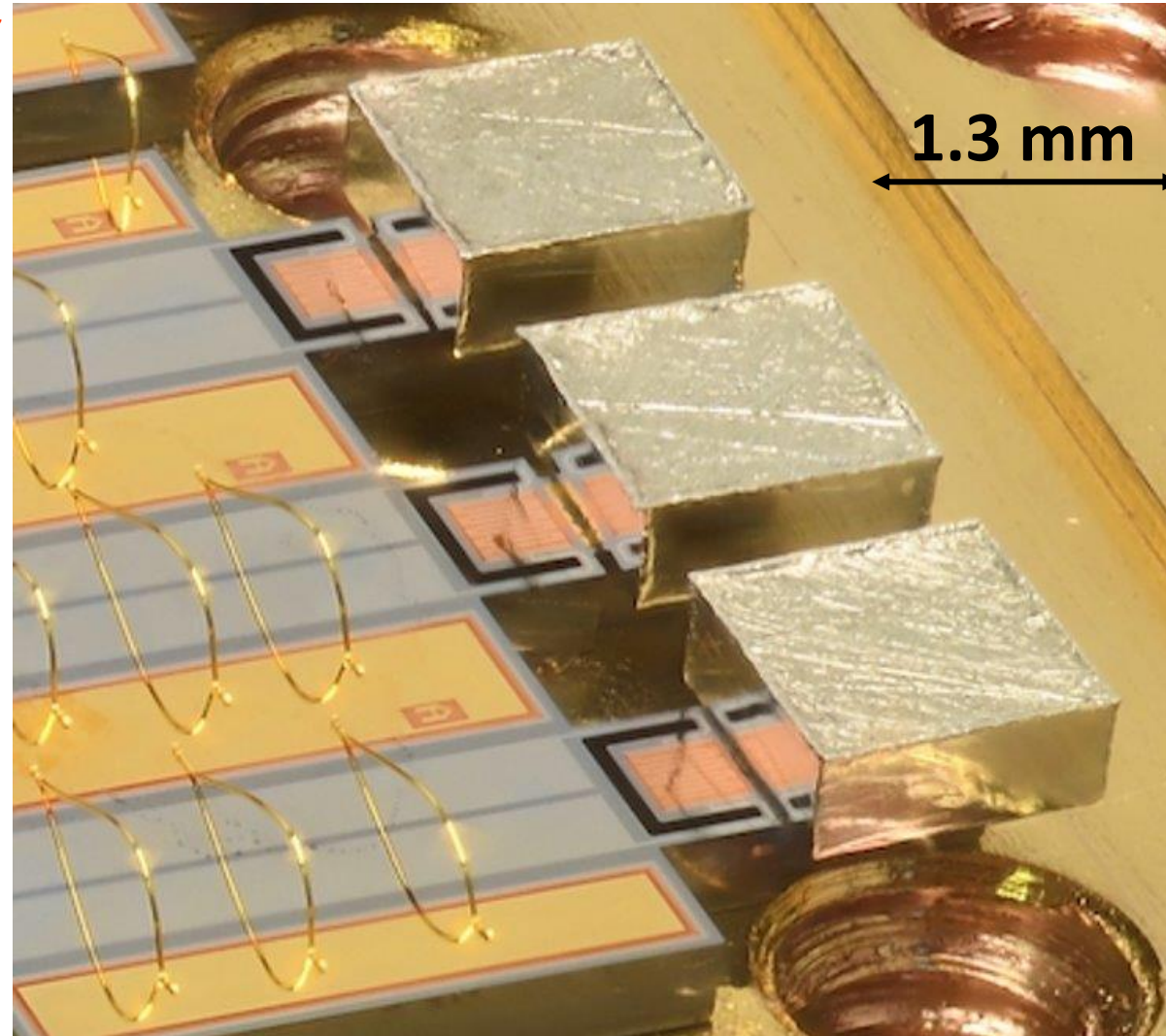
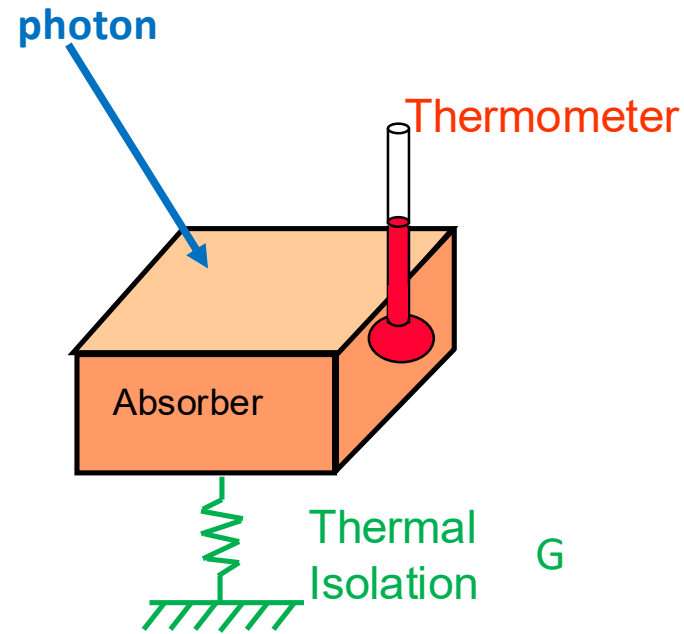
CURIE Detector Architecture: All-Silicon, No Membranes



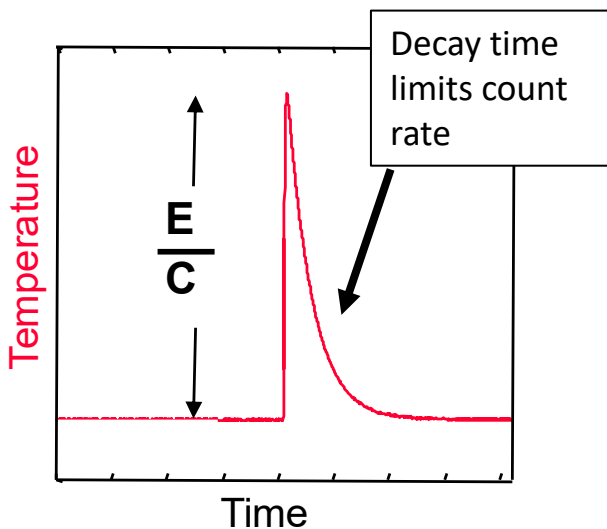
- ✓ Forging Assembly
- ✓ Physically Robust
- ✓ Access to wide range of G



CURIE Detector Architecture: All-Silicon, No Membranes



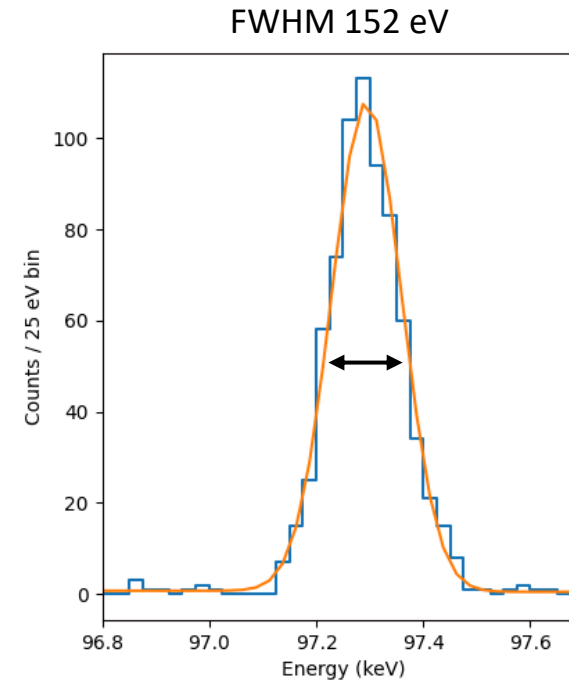
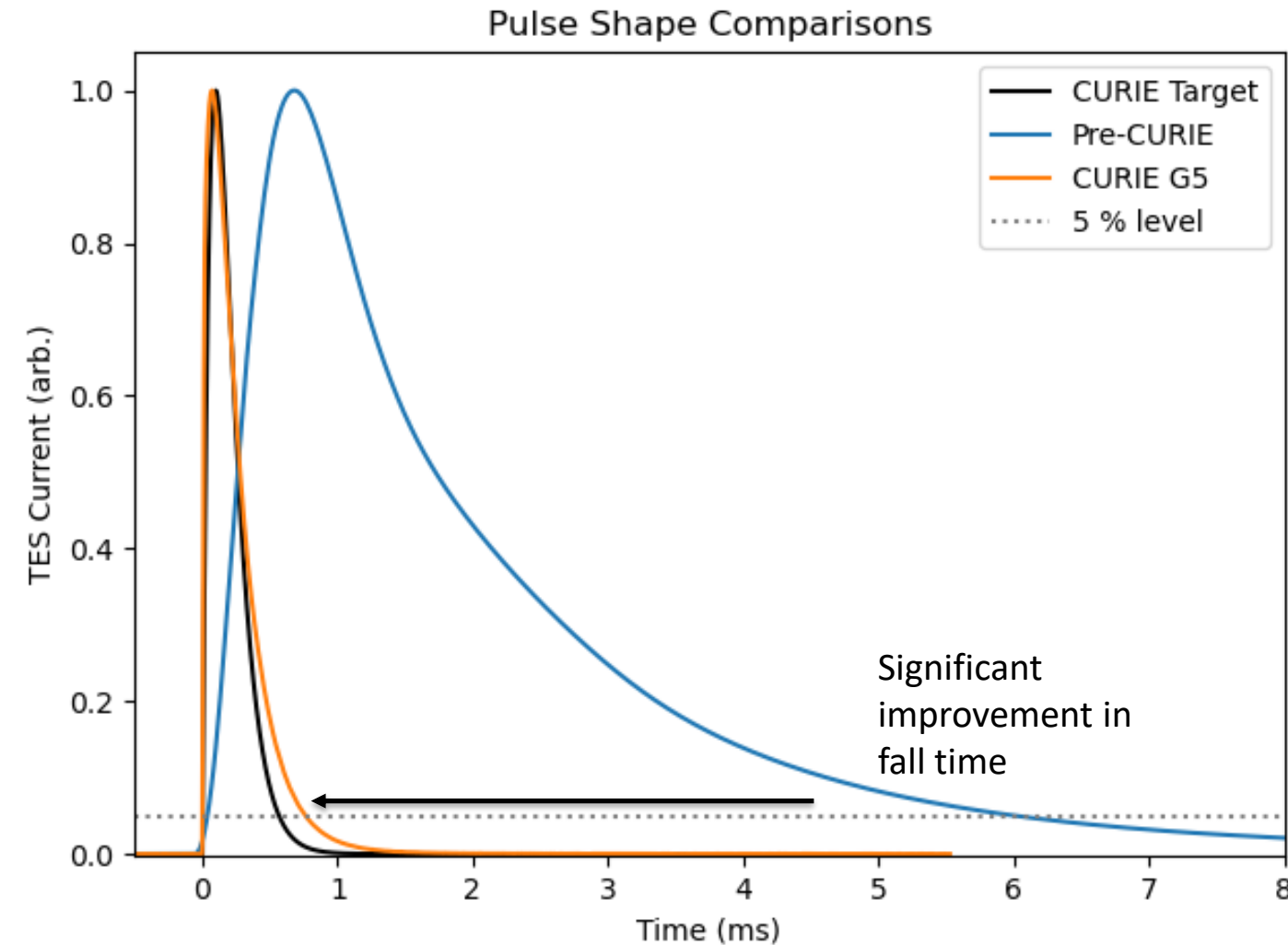
- ✓ Forging Assembly
- ✓ Physically Robust
- ✓ Access to wide range of G



Progress on Detector Speed And Energy Resolution

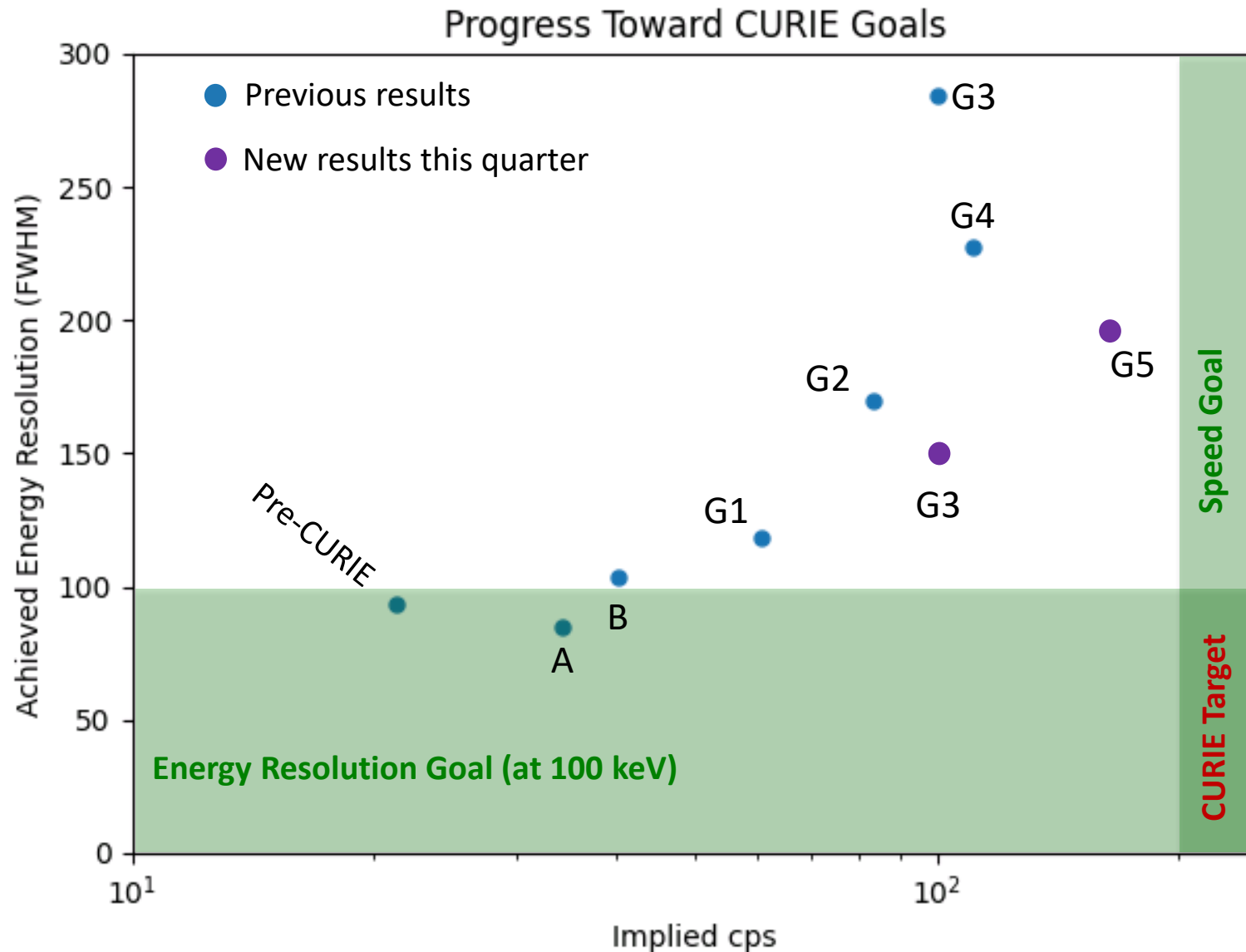
CURIE detector speed is compatible with
140 photons per second
→ within **1.4x** of target

Energy Resolution has degraded, but we
have a path to improvement: lower
operating temperatures (T)



$$\Delta E_{TES} \sim \sqrt{\frac{4k_B T^2 C}{\alpha}}$$

Summary of Progress

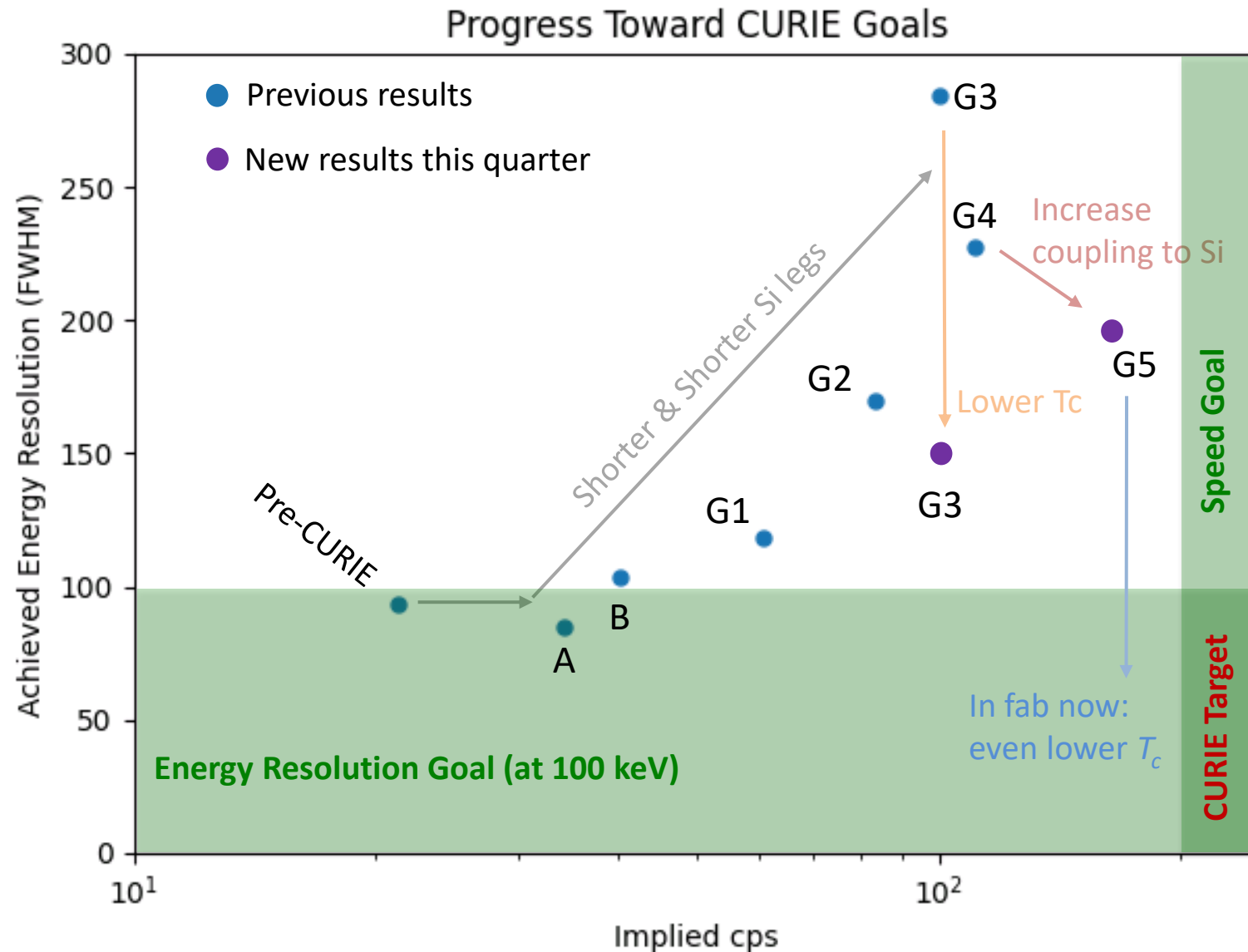


We are very close to target speed

Energy resolution needs to be improved, but there is a clear path via reducing operating temperature (superconducting critical temperature T_c)

Even 140 cps / detector is enormous step forward
→ aids commercialization

Summary of Progress



We are very close to target speed

Energy resolution needs to be improved, but there is a clear path via reducing operating temperature (superconducting critical temperature T_c)

Even 140 cps / detector is enormous step forward
→ aids commercialization

T2M: General Atomics shows interest in joining project as plus-up to accelerate deployment of fast next-generation online monitoring of nuclear materials

Current Scope

- ▶ Increase per-pixel count rate from 10-20 cps to 200 cps
- ▶ Deploy small (96-pixel) module at INL Analytical Lab

Long-term goal

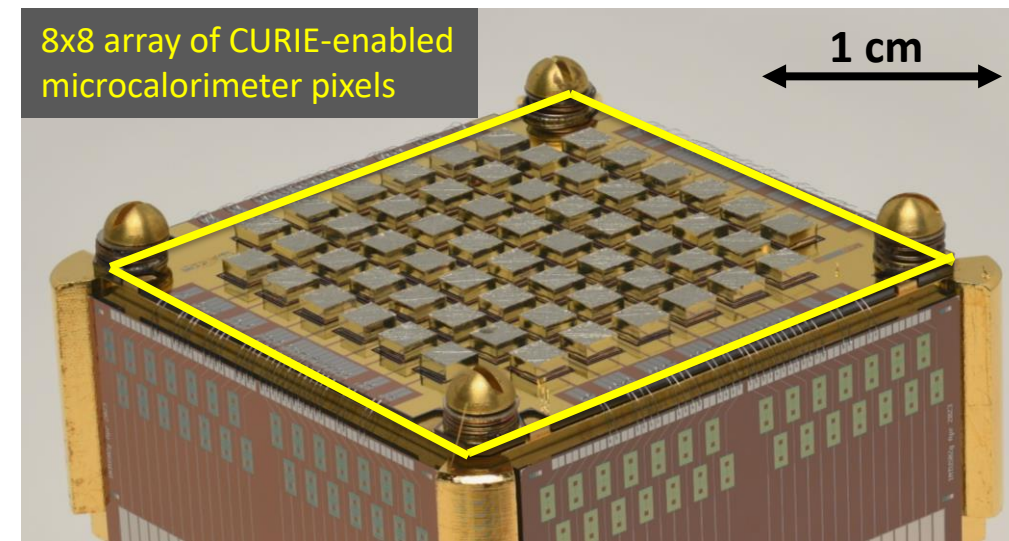
- ▶ Measurements in 2 minutes via 4000-pixel instrument

GA Plus-Up Technical Scope

- ▶ Solve absorber manufacture problem
- ▶ Build up GA microcalorimeter expertise

GA Plus-Up T2M Scope

- ▶ Evaluate commercial TES business opportunity
- ▶ Define requirements for product assembly



CURIE already leading to improved arrays of superconducting microcalorimeters

Summary + Next Steps



University of Colorado
Boulder

- New All-Si Gamma-Ray Microcalorimeter Architecture
- Already **14x** speed improvement
- Seeking plus-up to add General Atomics to project team

Spectrometer Metric	State of the Art	Proposed	Progress So Far
Energy Resolution	< 100 eV (at 100 keV)	unchanged	150 eV (improve via lower T)
Per-Detector Count Rates	10 – 20 cps	200 cps	140 cps
τ_1 % (time to fall to 1 % of peak)	16 ms	750 us	~ 1.5 ms
Athermal Tails	Significant	Negligible	Negligible



Next Steps

- Push forward to 20x speed improvement
- Bring energy resolution back to spec

